SafeShare AI

On-Device AI Privacy Guardian for Screen Sharing

Executive Summary

SafeShare AI is a local-first privacy assistant for screen sharing. It analyzes screenshots for potentially sensitive information and helps the user protect detected regions before sharing.

Problem Statement

People frequently share screens containing email addresses, phone numbers, account identifiers, API tokens, QR codes, faces and private documents. Manual checking is unreliable, especially when screen sharing starts quickly.

Proposed Solution

SafeShare AI scans a screenshot locally, detects supported sensitive patterns and visual elements, assigns a privacy-risk score, and produces a protected copy using blur, blackout or pixelation.

Snapdragon Strategy

The project is designed for Snapdragon-powered HP PCs. Qualcomm AI Hub supports computer-vision and multimodal model categories and exposes Snapdragon X Elite, X Plus 8-Core and X2 Elite targets, with ONNX Runtime and Qualcomm AI Runtime options. The final submission should include actual device measurements only after testing.

Innovation

The product connects detection to an immediate protection action. It does not stop at identifying sensitive content; it creates a protected screenshot and a CSV privacy report.

Architecture

Input screenshot -> local preprocessing -> OCR/vision detectors -> privacy rule engine -> risk score -> warning -> redaction -> protected output.

Features

Screenshot scanning; OCR-based PII patterns; face detection; QR detection; privacy score; annotations; blur/blackout/pixelation; CSV report; AI Hub/ONNX integration path.

Limitations and Validation

OCR and generic detectors can produce false positives or miss unusual content. NPU execution, runtime and performance must be verified on the actual target Snapdragon HP PC before being presented as completed results.

Future Scope

Live screen monitoring, custom models, enterprise policies, clipboard checks, browser integration and measured NPU acceleration.